ACC 317

MANAGEMENT INFORMATION SYSTEM (MIS)

LECTURE NOTE

CONCEPT OF MANAGEMENT INFORMATION SYSTEM

Introduction

Management: Management is art of getting things done through and with the people in formally organized groups. The basic functions performed by a manager in an organization are: Planning, controlling, staffing, organizing, and directing.

Information: Information is considered as valuable component of an organization. Information is data that is processed and is presented in a form which assists decision maker.

System: A system is defined as a set of elements which are joined together to achieve a common objective. The elements are interrelated and interdependent. Thus every system is said to be composed of subsystems. A system has one or multiple inputs, these inputs are processed through a transformation process to convert these input(s) to output.

Management Information Systems (MIS), referred to as Information Management and Systems, is the discipline covering the application of people, technologies, and procedures collectively called information systems, to solving business problems. MIS is a planned system of collecting, storing and disseminating data in the form of information needed to carry out the functions of management.

Management information system (MIS) provides information that organization require to manage themselves efficiently and effectively. Management information systems are typically computer systems used for managing the organizations. The five primary components of MIS are:

- Hardware
- Software
- Data (information for decision making),
- Procedures (design, development and documentation), and
- People (individuals, groups, or organizations)

Management information systems are distinct from other information systems because they are used to analyze and facilitate strategic and operational activities. Academically, the term is commonly used to refer to the study of how individuals, groups, and organizations evaluate, design, implement, manage, and utilize systems to generate information to improve efficiency and effectiveness of decision making, including systems termed decision support systems, expert systems, and executive information systems.

Data and Information

The terms data and information are often used synonymously. To computer professionals they are different.

Data: consists of facts and figures that are relatively meaningless to the user. It is a piece or collection of raw facts representing people, objects, and the day-today activities of an enterprise. An enterprise in this case can be a hospital, a school, a government ministry, a business organisation, etc.

Information: is data that have been put into a meaningful and useful context and communicated to a recipient who uses it to make decisions. Information consists of data, Images, text, documents, and voice, always organized in a meaningful context.

Information is one of the main types of resources that are available to the manager. Information can be managed just as any other resources. The information output of computers is used by managers, non-managers, and persons and organisation within the firm's environment.

Information is the bonding agent that holds an organisation together for better operation and coordination and survival and growth in an unfriendly competitive environment. Indeed, today's industries run on information.

A simple schematic that represents how we will think of information is shown in following figure.

Input (data) → Process (model) → Output (information)

Qualities/Attributes of Information

- Accessibility: This refers to the ease and speed with which information can be obtained.
- Comprehensiveness: This refers to the completeness of the information. Incomplete information affects management decisions that is, it leads to incomplete management action.
- Accuracy: This pertains to the degree of freedom from error of the information, especially when large volumes of data are involved. Transcription, transposition and computation error are examples.
- Appropriateness: This refers to how well the information relates to user's request. This information must be relevant to the matter at hand.
- Timeliness: This relates to the elapsed time of the cycle ie input, processing and reporting of output to users. For the timeliness of information to increase, the duration of this cycle must be reduced.
- Clarity: This refers to the degree to which information is free from ambiguity.
- Flexibility: This refers to the adaptability for use by more than one user.

Data Processing activities (Producing Information from Data)

Data processing consists of gathering the raw data input, evaluating and bringing order to it, and placing it in proper perspective so that useful information will be produced. All data processing whether done by hand or by the latest computer system, consists of three basic activities: capturing the input data, manipulating the data, and managing the output results.

- Capturing: Data must be originated in some form and verified for accuracy prior to further processing. They may initially be recorded on paper source documents and then converted into a machine usable form for processing, or they may be captured directly in a paperless machine-readable form.
- Verifying: Refers to the checking or validating of data to ensure that it was captured and recorded correctly. Examples are; a person reviewing another's work, the use of check digits in coding structures etc.
- Manipulating the data: One or more of the following operations may then have to be performed on the gathered data:
  - Classifying: It places data element into specific categories which provide meaning for the user e.g. Sales data can be classified as inventory type, size, customer, etc.
  - Arranging (sorting): Places data element in a specified or predetermined sequence, e.g. an inventory file can be arranged by product code, activity level, Naira value or by ant other attribute coded in the file and deemed desirable by a user.
  - Summarizing: Combines or aggregates data element, e.g. reducing data in the logical sense, as in the example where personnel manager wants a list of names of employees assigned to a department in a form.
  - Calculating: Entails the arithmetic and/or logical manipulation of data e.g. computation to derive employees pay, customer's bill, and students' grade point averages. Sophisticated calculations include linear programming, fore-casting etc.

Information Systems

An information system is a combination of software, hardware, and telecommunication networks to collect useful data, especially in an organisation. Many businesses use information technology to complete and manage their operations, interact with their consumers, and stay ahead of their competition.

Types of Information Systems

In early day of business computing, computers were usually used to replace the existing manual systems. But , in this modern day ,managers make decisions to solve problems, and information is used in making the decisions . Information is presented in both oral and written forms by an information processor. The computer portion of the information processor contains each of the following computer-based application areas:

- √Transaction Processing Systems (TPS): This is a system that processes data arising from the occurrence of a firm's business transactions such as Accounts receivable, Payroll, Inventory Control, Hotel reservation system, Sales order entry, Accounts payable and others. Transaction processing systems are computerized systems that perform and record the daily transactions necessary to the conduct of business. They serve the operational level of business organization hierarchy .These systems include both the accurate recording of transactions as well as the control procedures used in issuing such documents as invoices, customer statements etc.
- √Information Reporting Systems (IRS): is an information system that provides predefined types of information to management for relatively structured types of decisions. The main output of the IRS is displayed on screens either in continuous scrolled or static form. IRS are used for both management planning and management control functions.
- √Decision Support Systems: A decision support system is a form of information system that provides tools that help managers make decisions with both internal and information in the manner that best suits the decision they are currently trying to make. The DSS provides the manager with computing and communications capabilities to develop his or her own decision models and information banks. A DSS is not intended to make decisions for managers, but rather to provide them with a set of capabilities to enable them to generate the information they feel they need to make decisions that are semi structured, unique or rigidly changing and not easily specified in advance unlike TPS.
- √Executive information systems: The term executive is often used to describe a manager on the strategic planning level. An executive support systems (ESS) is also known as an executive information system (EIS). It is a decision support system that is designed to meet the special needs of executives. It is a system that provides information to the executive on the overall performance of the firm.
- √Artificial Intelligence or Expert Systems: Artificial Intelligence is the activity of providing such machines as computers with the ability to display behaviour that would be regarded as intelligent if it were observed in human; that is the ability that can be imparted to computers to enable them display intelligent, humanlike behaviour.

Real time and Online.

In this technology world, we all are using online systems as well as real-time systems, both of these systems are in use for different tasks but sometimes people are confused between the two so let's understand these in detail and also have a look at advantages and their limitations.

Online Systems

Online systems are the type of systems that are logged in or connected to the internet, such systems include online gaming platforms, chatting through different social media platforms, etc. A number of applications are available which are online in nature.

Advantages of Online Systems:

- Online systems help in making communication possible in the whole world at a faster speed.
- Online systems are less costly when compared to real-time systems, anyone can use them with the help of the internet.
- Response time is really good in the case of many online systems like booking systems or online shopping sites.
- These systems do not have heavy hardware requirements means they can be used with simple hardware resources.

Limitations of Online Systems:

- There are times when a large number of data is fetched or either sent to online systems and in such cases, these systems fail to handle that data.
- They are not providing in-time information means these systems lag when compared to real-time systems hence these systems cannot be used for high-priority tasks.

Real-Time Systems

Real-time systems are those systems that do not lag and respond instantly. These are different from online systems in the sense that they transform their state with correspondence to real or physical time.

Advantages of Real-Time Systems:

- These are the fastest available systems and hence are used in top priority operations like rocket launching systems, military communications, etc.
- The response time of these systems is instant, they don't lag and provide real-time information or services.
- Real-time systems are the foundation of advanced technology in various fields like communication, information retrieval, etc.
- These systems are able to handle a huge amount of data.

Limitations of Real-Time Systems:

- These systems are expensive so these are not available for use to everybody.
- Heavy system requirement to use these systems, these systems need special software and hardware resources.

Difference between Online and Real-Time Systems

The following table differentiates Online Systems and Real-time systems

S/N	Online System	Real-Time Systems
1.	These are the type of systems that works by connecting to the internet to update or share information.	These are the systems that are priority-based and include a sense of real physical time to pass information from one system to another.
2.	There is some lag in these systems.	These are instant without any lag.
3.	All online systems are not real-time.	All real-time systems are online.
4.	It is used for less important functions to achieve.	These systems are of high importance so there are limits on latency.
5.	Online systems are made for general purposes like communication, gaming, etc.	These are made for specific purposes like systems used in defense and national securities.
6.	System implementation is simple and can be easily designed.	System implementation is complex and designing such systems is a difficult task.
7.	The performance of such systems is low.	The performance of such systems is high.
8.	Sending an email is an example of the online system.	Automatic electronic systems are real-time systems.

Time sharing

Time-sharing is the name given to the computer technology established to provide on-line access to a large computer through remote terminals and better utilization of equipment by having more than one user share a computer. In data processing, time sharing is a method of operation in which multiple users with different programs interact nearly simultaneously with the central processing unit (CPU) of a large-scale digital computer.

CONCEPT OF SOFTWARE SYSTEM

Introduction

System software refers to the low-level software that manages and controls a computer's hardware and provides basic services to higher-level software. System software is software that provides a platform for other software. Some examples can be operating systems, antivirus software, disk formatting software, Computer language translators, etc. These are commonly prepared by computer manufacturers. This software consists of programs written in low-level languages, used to interact with the hardware at a very basic level. System software serves as the interface between the hardware and the end users.

Advantages of System Software:

- Resource management: System software manages and allocates resources such as memory, CPU, and input/output devices to different programs.
- Improved performance: System software optimizes the performance of the computer and reduces the workload on the user.
- Security: System software provides security features such as firewalls, anti-virus protection, and access controls to protect the computer from malicious attacks.
- Compatibility: System software ensures compatibility between different hardware and software components, making it easier for users to work with a wide range of devices and software.
- Ease of use: System software provides a user-friendly interface and graphical environment, making it easier for users to interact with and control the computer.
- Reliability: System software helps ensure the stability and reliability of the computer, reducing the risk of crashes and malfunctions.
- Increased functionality: System software provides a range of tools and utilities for performing various tasks, increasing the functionality and versatility of the computer.

Disadvantages of System Software:

- Complexity: System software can be complex and difficult to understand, especially for non-technical users.
- Cost: Some system software, such as operating systems and security software, can be expensive.
- System Overhead: The use of system software can result in increased system overhead, which can slow down the performance of the computer and reduce its efficiency.
- Vulnerability: System software, especially the operating system, can be vulnerable to security threats and viruses, which can compromise the security and stability of the computer.
- Upgrades: Upgrading to a newer version of system software can be time-consuming and may cause compatibility issues with existing software and hardware.
- Limited Customizability: Some system software may have limited options for customization, making it difficult for users to personalize their computing experience.
- Dependency: Other software programs and devices may depend on the system software, making it difficult to replace or upgrade without disrupting other systems.

Types of Software

There are two (2) main types of software:

- The system program
- The application program (package)

The System Program: The system programs are those programs written by the computer manufacturers. These programs are supplied with the computer system on purchase. The system programs includes the operating system, the language translator and the utility/services program and library program.

The operating system: - The operating system are those system program that interferes between the user of the computer and the computer hardware. The operating system is described as the manager of computer resources. It is the house keeper of the computer resources, and as the housekeeper, it knows the condition and the location of each resources. It allows the computer user to make use of those resources and reclaims them from the user after use the safe keep them again for future use.

An operating system (OS) is a type of system software that manages a computer's hardware and software resources. It provides common services for computer programs. An OS acts as a link between the software and the hardware. It controls and keeps a record of the execution of all other programs that are present in the computer, including application programs and other system software.

The most important tasks performed by the operating system are

- Memory Management: The OS keeps track of the primary memory and allocates the memory when a process requests it.
- Processor Management: Allocates the main memory (RAM) to a process and de-allocates it when it is no longer required.
- File Management: Allocates and de-allocates the resources and decides who gets the resources.
- Security: Prevents unauthorized access to programs and data using passwords.
- Error-detecting Aids: Production of dumps, traces, error messages, and other debugging and error-detecting methods.
- Scheduling: The OS schedules the process through its scheduling algorithms.

Language Translator:

- The Assembler: The assembler is the language translator that converts programs written in assembly language to machine language. It translates the assembly programs contained in the source file to the target or object language which the computer can now execute if it is error free. Examples are: Microsoft MACRO ASSEMBLER, Microsoft Quick Assembler, A86 Macro Assembler.
- The Compiler: The complier is a language translator that converts programs from high level language to machine language. It picks the high level language contained in the source file, converts all to machine language and create a new file called object file.
- The Interpreter: The interpreter is a language translator that translates the high level language to machine language. It pick the statements line by line and translates them one statements after the other sequentially. Any line translated, is executed by the computer before the next line is translated.

The Utility programs: The utility programs just like the name sounds is the type of system programs that are utilized by the users of computer to solve one data processing operation or the other. They remain resident in the storage media where they are stored until they are invoiced through a command to perform. Examples of these programs are: copy, format, sorting, dumping etc.

Application Packages: This is a specially designed program written by the professional programmers for solving specific data processing needs, either commercial or scientific. A package is a complete suite of ready-made programs with its documentation covering a processing routine. It is usually supplied by a software house or manufacturer. The supply might either be on purchase or on lease. It is usually intended to meet the needs of a wide range of user companies. Examples of such applications include: Payrolls, auditing, Stock control, Network Analysis, Dbase, Spreadsheet, WordPerfect, etc.

- Word processing: These are programs that turn a computer system into a powerful typewriting tool and more. It helps us to produce letters, reports, magazines, memos etc. Example of word processing programs are: WordStar, WordPerfect. PC Write, MultiMate, Ten Best and WordStar 2000.
- Database Management: Databases are the backbone of business computing. They store and retrieve information such as customer lists, inventories and notes. They can also sort information, letting users do things like search their customer list for customers in a given STATE. These programs help us to record and maintain information about people, places, things and management reports of all types. Examples are Dbase III & IV, FoxBASE, Oracle, FoxPro and Clipper, Microsoft Access, Paradox, Professional File, Windows File and Q&A etc.
- Accounting: Accounting programs keep track of business income and expenditures including accounts receivable, accounts payable, operating expenses, cash flow and payroll. Popular accounting programs include QuickBooks, DacEasy Accounting, Peachtree Accounting, Profit, One-Write, Paciolli 2000, and the ACCPAC series of programs.

- Spreadsheet: Spreadsheets are software packages that, more or less, turn a computer system into a sophisticated electronic calculator. Many spreadsheet packages also have presentation graphics generators, which take data and painlessly convert them into bar charts, line charts, and the like, for management presentations at meetings. These programs help us to produce financial reports and reports. Examples are Lotus 1-2-3, QuatoPro, SuperCalc, Multiplan, VP Planner, Planning Assistant and Microsoft Excel.
- Graphics Programs: Graphics programs create graphs and drawings that can be used in newsletters, posters, advertisements and other documents. Some allow users to import photographs into documents or create animated pictures for use in "multimedia" presentations. Popular graphics programs include PC Paintbrush, Professional Drave, IntelliDraw, illustrator, Persuasion, Cricket Draw, Corel Draw, Print Master, Print Shop, Microsoft PowerPoint, and Harvard Graphics.
- Desktop Publishing: Desktop publishing programs are extremely sophisticated word processors that can incorporate word processing and graphics files to create newsletters, manuals, files, advertisements, magazine and newspapers. Popular DTPs include Microsoft Express Publisher, Aldus PageMaker, Ventura Publisher and Quark XPress.
- Integrated Packages/All in One: Integrated programs are a program that combines the ability to do several general-purpose applications including three or four different functions , usually word processing, database, graphics and spreadsheet in one program. Some include communications software that, combined with a modem, will enable you to look up and dial telephone numbers easily. The most popular integrated packages include Microsoft office, WordPerfect Works, ClarisWorks, GeoWorks Pro and Lotus Symphony.
- Computer Aided Design: Computer Aided Design is a package that can be used to produce drawings of professional quality on paper as large as your printer or plotter can handle. Such drawings like engineering plans, models, maps, electrical circuit diagrams, planning diagrams, architectural and more are made possible by the use of computer. CAD is widely used in Industry, Commerce, and Education. Popular CAD Packages which can perform all these functions include like AutoCAD, Auto-Sketch etc.

Specific Purpose Application Program: These programs are designed and program to solve the specific problem by a computer specialist who is employed in the company. Examples include the following:

- Payroll, Stock control, Sales order processing and sales ledger, Purchase order processing and purchase ledger.

Those that are used for a specialized form of business e.g.

- Transaction processing banks, Airline seat reservation, Theatre seat reservation, Hotel room reservation, Point of sale software.

Stages involved in Programming.

Before computer program is successfully written, documented and installed, it must have passed through the following stages.

- Problem definition: Before any reasonable and meaningful program could be written, the problem that prompted it must have to be defined. No one solves a problem he does not know. The problem to be solved by computer should be well stated and understood before the solution will be worked out. From the solution, it is expected that the output of the problem is known and the input will be prepared to arrive at the output.
- Develop the algorithm: An algorithm is a well-defined set of instructions that is used to solve a particular problem in a finite number of steps. It involves unambiguous stating of the procedures and steps necessary to transform the input data into output. It possess a little difficulty to the program planner, and once accomplished successfully, the rest of the solution follows easily.
- Plan the logic of the program/flowcharting: The logic of the program will be planned using any of the program design tools it flowchart, pseudo code or hierarchy chart. The choice of the design tool used depends on the programmer, but the most popular and most handy is the use of the flowchart to organize the thought of the program planner and to check for any logic error or misrepresentation. A flowchart is a pictorial view of the program logic
- Write the computer program: After the design or planning the logic of the program using the flowchart, the next stage is the actual writing of the program using any of the programming languages in a proper sequence. This is called, coding of the program. This is done by strictly obeying the language syntax or following the established rules of the programming language.
- Type the program into computer: The next stage after writing the program, it to key the program into the computer. Any program that will be executed by the computer must be resident in the computer memory. The typing is generally made one line after the other.
- Test and debug the program: The moment the program has been keyed into the computer, the programmer is ready to see if the program is working. The program could be translated into machine language by either a compiler or interpreter depending on the language in use ie for BASIC program, when the command RUN is typed and entered, the program begins executing. If any rules language is broken, the program will not work. The errors must be removed before the program will start producing the output. Testing is very necessary to ensure that the correct and required answers are produced as the output.
- Document the work: Documentation helps the user to understand the program better. It identifies exactly the purpose of the program. It is always referred to whenever changes are to be made in the program to suite new development. It contains the following parts.
  - A statement of the problem
  - Algorithm and program plans (ie flowchart, hierarchy chart or pseudo code).
  - Description of input and output
  - Program listing
  - Test data and results
  - Technical details and instruction for the user.

Concept of Algorithm.

An algorithm is a set of instructions to obtain the solution of a given problem. Computer needs precise and well-defined instructions for finding solution of problems. If there is any ambiguity, the computer will not yield the right results. It is essential that all the stages of solution of a given problem be specified in details, correctly and clearly moreover, the steps must also be organized rightly so that a unique solution is obtained. A typical programming task can be divided into two phases:

Problem solving phase: In this stage an ordered sequence of steps that describe solution of the problem is produced. Their sequence of steps can be called anti-Algorithm

Implementation Phase: In this phase, the program is implemented in some programming languages. Algorithm may be set up for any type of problems, mathematical/scientific or business. Normally algorithms for mathematical and scientific problems involve mathematical formulas. Algorithms for business problems are generally descriptive and have little use of formula.

Features of an Algorithm,

- It should be simple
- It should be clear with no ambiguity
- It should head to unique solution of the problem
- It should involve a finite number of steps to arrive at a solution
- It should have the capability to handle unexpected situation.

Flowchart

Flowchart is a representation of the algorithm using standard symbols. Each symbols has a new function. The Algorithm steps determine which symbol to use to represent it in the flow each step is linked to another step by using the directional arrows.

A flowchart is a pictorial representation of an Algorithm or of the plan of solution of a problem. It indicates the process of solution, the relevant operations and computations, point of decision and other information that are part of the solution. Flowcharts are of particular importance for documenting a program. Special geometrical symbols are used to construct flowcharts. Each symbol represents an activity. The activity could be input/out of data, computation/processing of data, taking a decision, terminating the solution, etc. The symbols are joined by arrows to obtain a complete flowchart.

Uses of flowcharts

- It gives us an opportunity to see the entire system as a whole.
- It makes us to examine all possible logical outcomes in any process.
- It provides a tool for communicating i.e a flowchart helps to explain the system to others.
- To provide insight into alternative solutions.
- It allows us to see what will happen if we change the values of the variable in the system.

Introduction to Basic programming.

BASIC stands for Beginner's All-purpose Symbolic Instruction Code, and is a computer programming language that was invented in 1964. BASIC has the advantage of English-like commands that are easier to understand and remember than those of most other languages.

Basic was the first language made available for personal computers (Microsoft started its business selling a version) and in recent years it has returned to importance as VISUAL BASIC, though the latter bears little resemblance to earlier versions.

Visual Programming Language

Any language that uses the graphics or blocks that are already defined with the code and you just need to use those blocks without worrying about the lines of code is known as a visual programming language. In today's era majority of the programming languages are text-based i.e. we have to write the lines of code to perform the specific task like in C or C++.

Applications of Visual Programming language:

VPL can be used in multiple domains like multimedia, educational purpose, video games, automation. Let's see them in brief:

- Multimedia:- VPL helps users create multimedia without worrying about the real code or other complex features. It narrows down to specific functions and with the help of those functions, multimedia is created.
- Educational Purpose:-Scratch VPL, etc are used to help students in their projects and make them familiar with the coding.
- Videogames:-VPL helps to create the videogames without writing lines of codes Ex- Scratch VPL is used to make videogames.

Advantages of visual programming language:

- Easy to convert ideas into reality for example you don't know how to code so you can start with VPL and then switch to actual coding.
- Visuals are easy to grasp i.e. to develop something in visual programming language requires less efforts.
- It includes a variety of built in objects that are required while creating something using VPL.
- It is a beginner-friendly also anyone will be able to derive the logic without worrying about writing lines of code.
- Adding a user-specific code is also available and simple as it allows to create of blocks as per the convenience of the user.

Disadvantages of visual programming language:

- These languages require more memory as they use graphics, as a result, their execution is also slow and a large amount of memory is occupied by them.
- They can only work in an operating system like windows, mac, or any other operating system which supports graphics.
- As the inbuilt functions are not sufficient so you have to add your custom code as a result it is cumbersome.
- Only limited functions are present in these languages.
- Adding our custom code as a block requires coding knowledge or else you have to work with limited functions which are provided with the language.

System Development Circle

Most IT projects work in cycles. First, the needs of the computer users must be analyzed. This task is often performed by a professional Systems Analysts who will ask the users exactly what they would like the system to do, and then draw up plans on how this can be implemented on a real computer based system. The programmer will take the specifications from the Systems Analyst and then convert the broad brushstrokes into actual computer programs. Ideally at this point there should be testing and input from the users so that what is produced by the programmers is actually what they asked for.

Finally, there is the implementation process during which all users are introduced to the new systems, which often involves an element of training. Once the users start using the new system, they will often suggest new improvements and the whole process is started all over again. These are methodologies for defining a systems development cycle and often you will see four key stages, as listed below.

Feasibility Study
Design
Programming
Implementation

Feasibility Study

The main objective of a feasibility study is to test the technical, social and economic feasibility of developing a system. This is done by investigating the existing system in the area under investigation and generating ideas about a new system. The proposed system(s) must be evaluated from a technical viewpoint first, and if technically feasible their impact on the organisation and staff must be assessed. If compatible social and technical systems can be devised, then they must be tested for economic feasibility.

Assessing technical feasibility: The assessment of technical feasibility must be based on an outline design of system requirements in terms of inputs, outputs, files, programs, procedures and staff. This can then be quantified in terms of volumes of data, trends, frequency of updating, cycles of activity, etc., in order to give an indication of the scale of the technical system. Methods used for investigation, analysis and design of this system are described in the chapters which follow. Having identified an outline system, the investigator must go on to suggest the type of equipment required, methods of developing the system, and methods of running the system once it has been designed.

Assessing social feasibility: The assessment of social feasibility will be done alongside technical feasibility. Each of the alternative technical solutions which emerge must be evaluated for its social implications. The needs of various people affected by the proposed system (both directly and indirectly) must be taken into account. Impact on organisation structure, authority, salary levels, group relationships, and jobs should be considered-not just in negative terms (i.e. bad effects) but positively (i.e. what can be improved? what contribution can be made?).

Assessing economic feasibility: Proposed or developing systems must be justified by cost and benefit criteria to entire that effort is concentrated on projects which will give the best return at the earliest opportunity. The technique of cost benefit analysis is often used as a basis for assessing economic feasibility. This type of analysis is not new; it is carried out as a matter of course for many other capital development projects. The factors for evaluation are:

- Cost of operation of the existing and proposed system;
- Cost of development of the proposed system;
- The benefits of the proposed system.

System Analysis and Design.

System Analysis and System Design are two stages of the software development life cycle. System Analysis is a process of collecting and analyzing the requirements of the system whereas System Design is a process of creating a design for the system to meet the requirements. Both are important stages as it helps to create an effective system with all the features and functions.

What is System Analysis?

System Analysis is a process of understanding the system requirements and its environment. It is one of the initial stages of the software development life cycle. System analysis is the process of breaking the system down into its individual components and understanding how each component interacts with the other components to accomplish the system's overall goal. In this process, the analyst collects the requirements of the system and documents them.

Characteristics of system analysis
- It is the study of the existing system to identify the problem areas.
- It is a process of understanding the system requirements and its environment.
- It involves gathering and understanding the user's requirements.
- It involves analyzing the system in terms of its current and future needs.

Advantages of system analysis
- It helps to identify the problems and their causes.
- It helps to understand the functional and non-functional requirements of the system.
- It helps to develop better solutions.
- It helps identify the areas of improvement.

Limitations of system analysis
- It can be time-consuming.
- It can be costly.
- It can be difficult to get accurate information.

What is System Design?

System Design is the process of creating a design for the system to meet the requirements. System design is the process of designing the architecture, components, modules, interfaces, and data for a system to satisfy the specified requirements. It involves the design of the system architecture, components, modules, interfaces, and data.

Characteristics of system design
- It is the process of creating a design for the system.
- It involves the design of the system architecture, components, modules, interfaces, and data.
- It involves identifying the modules and components of the system.
- It involves creating the user interface and database design.

Advantages of system design
- It helps to create an efficient system.
- It helps identify the areas of improvement.
- It helps to reduce the development cost.
- It helps to create a better user experience.

Limitations of system design
- It can be time-consuming.
- It can be costly.
- It can be difficult to get accurate information.

System implementation and documentation.

Systems implementation is the process of:
- Defining how the information system should be built (i.e., physical system design),
- Ensuring that the information system is operational and used,
- Ensuring that the information system meets quality standard (i.e., quality assurance).

Implementation issues
- Testing
- Documentation

Testing process
- Program testing with test data
- Link testing with test data
- Full systems testing with test data (alpha test)
- Full systems testing with live data (beta test)

Testing guidelines

- Test different aspects of the system, e.g., response time, response to boundary data, response to no input, response to heavy volumes of input
- Test anything that could go wrong or be wrong about a system
- Test the most frequently used parts of the system at a minimum
- Use debugging tools, e.g., symbolic debugger

Program documentations

This is the act of keeping/maintaining all materials that serve primarily to describe a system/program and make it more readily understandable rather than to contribute in some way to actual operation of the system. Documentation is frequently classified according to purpose; thus for a given system there may be requirements documents, design document, and so on.

Why is program documentation important?

- The main purpose of program documentation is to describe the design of your program. The documentation also provides the framework in which to place the code as coding progresses, the code is inserted into the framework already created by the program documentation. .
- Documentation is important to tell other programmers what the program does and how it works. In the "real world" and in some classes here at BGSU, programmers often work in teams to develop code. Documentation helps others on the team to understand your work.
- Maintenance and debugging are needed sooner or later for most programs and these are frequently done by someone other than the original programmer. Documentation can help the programmer who is making the modifications understand your code.
- Documenting your program during development helps you to maintain your sanity.

Types of Software Documentation:

- Requirement Documentation: It is the description of how the software shall perform and which environment setup would be appropriate to have the best out of it. These are generated while the software is under development and is supplied to the tester groups too.
- Architectural Documentation: Architecture documentation is a special type of documentation that concerns the design. It contains very little code and is more focused on the components of the system, their roles, and working. It also shows the data flow throughout the system.
- Technical Documentation: These contain the technical aspects of the software like API, algorithms, etc.
- End-user Documentation: As the name suggests these are made for the end user. It contains support resources for the end user.

Advantages of Software Documentation:

- The presence of documentation helps in keeping the track of all aspects of an application and also improves the quality of the software product.
- The main focus is based on the development, maintenance, and knowledge transfer to other developers.
- Helps development teams during development.
- Helps end-users in using the product.
- Improves overall quality of software product
- It cuts down duplicative work.
- Makes easier to understand code.

Disadvantages of software documentation:

- The documenting code is time-consuming.
- The software development process often takes place under time pressure, due to which many times the documentation updates don't match the updated code.
- The documentation has no influence on the performance of an application.
- Documenting is not so fun, it's sometimes boring to a certain extent.

Program Maintenance

Program/software maintenance is the modification of a software product after delivery to correct faults, to improve performance or other attributes, or to adapt the product to a modified environment. Once the Program is made, it is saved and stored as a software package. After some time if the program needs improvement or modification, the saved program is modified and saves effort and time as programs need not to made from the very beginning. Hence Modifications will meet the purpose only. Program maintenance is done as:

- Keep the last program.
- Modify the program for required improvements.
- Do not make new program from scratch, just use the last saved program.
- Save time and effort.

Database Management

A database management system (DBMS) is a computer software application that interacts with the user, other applications, and the database itself to capture and analyze data. A general-purpose DBMS is designed to allow the definition, creation, querying, update, and administration of databases.

Types of Database software

Access, DataCom/DB, DB2, FoxPro, IDMS, Informix, Ingres, Oracle, Paradox, SQL Server Etc.

Purpose of Database Management Systems

Organizations use large amounts of data. A database management system (DBMS) is a software tool that makes it possible to organize data in a database. The ultimate purpose of a database management system is to store and transform data into information to support making decisions.

A DBMS consists of the following three elements:

- The physical database is the collection of files that contain the data
- The database engine is the software that makes it possible to access and modify the contents of the database
- The database scheme is the specification of the logical structure of the data stored in the database

Functions of a DBMS

DBMS provides the following functions:

- Concurrency: DBMS provides concurrent access (meaning 'at the same time') to the same database by multiple users
- Security: DBMS provides security rules to determine access rights of users
- Backup and recovery: DBMS processes to back-up the data regularly and recover data if a problem occurs
- Integrity: database structure and rules improve the integrity of the data
- Data descriptions: a data dictionary provides a description of the data

Benefits of DBMS.

There are a number of benefits to using a DBMS.

- A DBMS provides automated methods to create, store and retrieve data. It may take some time to set up these methods, but once in place, a DBMS can make tedious manual tasks a thing of the past.
- A DBMS reduces data redundancy and inconsistency. Have you ever had different versions of the same file on your computer hard drive? The same thing happens in organizations. A well-designed DBMS will eliminate redundancy.
- A DBMS allows for concurrent access by multiple users, each with their own specific role. Some users only need to view the data, some contribute to adding new data, while others design and manage the database - all at the same time!
- Others are: Crash recovery, Reduced application development time, Providing persistent storage for program objects

INTRODUCTION OF INTERNET

Introduction

A network is a group of two or more computer systems (Multiple gadgets, additionally called hosts), which are related through a couple of channels for the motive of sending and receiving data (records/media) in a shared environment. The community also can consist of several gadgets/mediums that resource communique among or extra machines; those gadgets are called Network devices and consist of routers, switches, hubs, and bridges, amongst others.

Internet is a worldwide interconnected network of hundreds of thousands of computers of various types that belong to multiple networks. The Internet gives a huge variety of statistics and communique offerings, which includes forums, databases, email, and hypertext. It is made of the neighborhood to global personal, public networks connected through plenty of digital, wireless, and networking technologies.

Working with the internet addressing

The internet is a global computer network that connects various devices and sends a lot of information and media. It uses an Internet Protocol (IP) and Transport Control Protocol (TCP)-based packet routing network. TCP and IP work together to ensure that data transmission across the internet is consistent and reliable, regardless of the device or location. Data is delivered across the internet in the form of messages and packets. A message is a piece of data delivered over the internet, but before it is sent, it is broken down into smaller pieces known as packets.

IP is a set of rules that control how data is transmitted from one computer to another via the internet. The IP system receives further instructions on how the data should be transferred using a numerical address (IP Address). The TCP is used with IP to ensure that data is transferred in a secure and reliable manner. This ensures that no packets are lost, that packets are reassembled in the correct order, and that there is no delay that degrades data quality.

Uses of the Internet:

- E-mail: E-mail is an electronic message sent across a network from one computer user to one or more recipients. It refers to the internet services in which messages are sent from and received by servers.
- Web Chat: Web chat is an application that allows you to send and receive messages in real-time with others. By using Internet chat software, chat software is interactive software that allows users to enter comments in one window and receive responses from others who are using the same software in another window.
- World Wide Web: The World Wide Web is the Internet's most popular information exchange service. It provides users with access to a large number of documents that are linked together using hypertext or hyperlinks.
- E-commerce: E-commerce refers to electronic business transactions made over the Internet. It encompasses a wide range of product and service-related online business activities.
- Internet telephony: The technique that converts analog speech impulses into digital signals and routes them through packet-switched networks of the internet is known as internet telephony.
- Video conferencing: The term "video conferencing" refers to the use of voice and images to communicate amongst users.

Differentiation between Network and Internet

The number one distinction between a network and the internet is that a network is made of computer systems that are bodily related and may be used as a personal laptop at the same time as additionally sharing records. The Internet, on the alternative hand, might be an era that connects those small and massive networks and creates a brand new in-intensity community.

Advantages of the Internet:

- It is the best source of a wide range of information. There is no better place to conduct research than the internet.
- Online gaming, talking, browsing, music, movies, dramas, and TV series are quickly becoming the most popular ways to pass the time.

- Because there are hundreds of thousands of newsgroups and services that keep you updated with every tick of the clock, the Internet is a source of the most recent news.
- Because of virtual shops where you may buy anything you want and need without leaving your house, internet shopping is becoming increasingly popular. Recently, virtual shops have been making a lot of money.
- With the emergence of online businesses, virtual stores, and credit card usage, purchasing goods without going to the store has never been easier.

Disadvantages of the Internet:

- Spending too much time on the internet is hazardous for the young generation's physical and mental health.
- Children who use the internet develop an addiction, which is quite dangerous.
- It is now quite easy to decipher someone's chat or email messages thanks to the hacking community.
- With the emergence of online stores, people prefer to order online rather than going to local stores which results in less social interactions among people.

Networks Access.

An access network is a type of network which physically connects an end system to the immediate router (also known as the "edge router") on a path from the end system to any other distant end system. Examples of access networks are ISP, home networks, enterprise networks, ADSL, mobile network, FTTH etc.

Types of access networks:

- Ethernet – It is the most commonly installed wired LAN technology and it provides services on the Physical and Data Link Layer of OSI reference model. Ethernet LAN typically uses coaxial cable or twisted pair wires.
- DSL – DSL stands for Digital Subscriber Line and DSL brings a connection into your home through telephone lines and the line can carry both data and voice signals and the data part of the line is continuously connected. With DSL you can able to use the Internet and make phone calls simultaneously. DSL modem uses the telephone lines to exchange data with digital subscriber line access multiplexer (DSLAMs). In DSL we get 24 Mbps downstream and 2.5 Mbps upstream.
- FTTH – Fiber to the home (FTTH) uses optical fiber from a Central Office (CO) directly to individual buildings and it provides high-speed Internet access among all access networks. It ensures high initial investment but lesser future investment and it is the most expensive and most future-proof option amongst all these access networks.
- Wireless LANs – It links two or more devices using wireless communication within a range. It uses high-frequency radio waves and often include an access point for connecting to the Internet.
- 3G and LTE – It uses cellular telephony to send or receive packets through a nearby base station operated by the cellular network provider. The term "3G internet" refers to the third generation of mobile phone standards as set by the International Telecommunications Union (ITU). Long Term Evolution (LTE) offers high-speed wireless communication for mobile devices and increased network capacity.
- Hybrid Fiber Coaxial (HFC) – HFC is a combination of fiber optic and coaxial cable that is widely used by cable television operators to provide high-speed internet access. The fiber optic cable is used to connect the head end to the neighborhood, while the coaxial cable is used to connect individual homes to the network.
- Satellite Internet – Satellite internet is a wireless connection that uses satellite communication to deliver internet access to remote and rural areas. It has a higher latency and lower bandwidth compared to other access networks, but it can provide internet access in areas where other options are not available.
- Power Line Communication (PLC) – PLC uses the existing electrical wiring in a building to transmit data signals. It is a low-cost alternative to traditional wired networks and can be used to provide internet access in buildings where it is difficult to install new cables.
- WiMAX – WiMAX (Worldwide Interoperability for Microwave Access) is a wireless access network technology that provides high-speed internet access over a wide area. It is commonly used in rural and suburban areas where it is difficult or expensive to deploy wired networks.
- 5G – 5G is the latest wireless communication technology that offers high-speed internet access and increased network capacity. It is designed to support a wide range of applications, including virtual and augmented reality, autonomous vehicles, and smart cities. 5G networks are being rolled out globally, and they are expected to transform the way we connect to the internet.